Credit Risk Modeling - Logistic Regression

Keyword:	PD Model ROC-AUC: 0.86
Skills used:	Logistic Regression

This project implements a Probability of Default (PD) model using Logistic Regression on the Lending Club dataset. Under the IRB (Internal Ratings-Based) approach, banks are permitted to develop their own models to estimate borrower PD using internal historical data, enabling more efficient capital allocation and a competitive advantage over the standardized baseline.

Logistic regression remains the industry standard for credit risk modeling. While tree-based methods (Random Forest, XGBoost) can achieve higher classification performance, logistic regression is preferred for its interpretability, which is a regulatory requirement under the Basel framework.

1. Preprocessing Data

→ ***Handle Large Number of Features:*** The Lending Club Dataset contains 2260668 rows and 145 columns, which is a very large amount of features. To process this, I divided it into 3 data types:

- Categorical (dtype = object)
- Numerical (dtype = numeric or float)
- Datetime (dtype = object but had the suffix _d or _date in column name)

and processed each data type separately.

∗General cleaning:

→ ***Near Zero-variance Problem:*** *A feature with a dominant value accounting for the majority of the data. Some columns have mostly missing value or a strongly dominant value (>99% of data) like total_rec_late_fee 99% are zeros and will not give much predictive power. I decided to also drop column where >90% data are null because it's also a type of near-zero variance (with a dominant value which is null, so even if we impute, it's still the same problem).*

1.1 Preprocess Categorical Data

→ ***High Cardinality Problem:*** *A feature with high cardinality mean it has too many unique values and unlikely to give predictive signals. It also gives too many variables when we later use on-hot-encoding. So I filter out variables like emp_title, desc (loan description) , title (purpose description). These variables contain free-text, cardinality of >500k (having 500k+ unique values) and requires NLP techniques to process.*

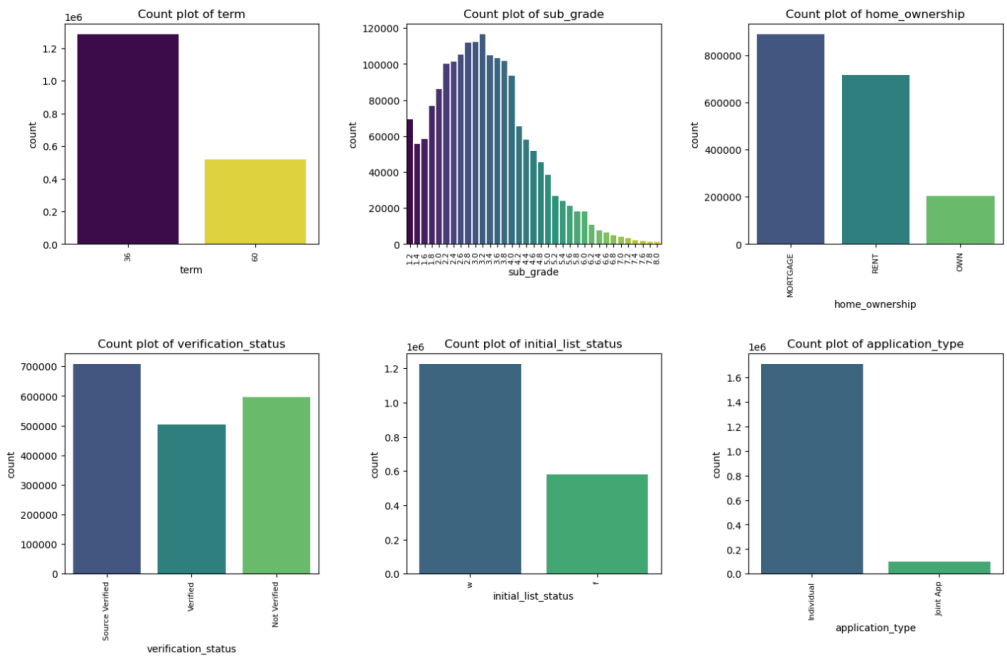
For other categorical columns, I used Weight of Evidence (WoE) (with adjustment to avoid divide by zero) and Information Value (IV) with Coarse Classing to judge which feature contribute meaningfully to separating the labels.

I used the benchmark 0.02 < IV < 1, which mean features with normal - strong predictive signal in industry standard to select the features.

$$Information\ Value = \ln\left(\frac{\%good}{\%bad}\right) * (\%good - \%bad)$$

$$Weight\ Of\ Evidence\ = \ln\left(\frac{\%good}{\%bad}\right)$$

Those that I kept are: ['term', 'home_ownership', 'verification_status', 'initial_list_status', 'application_type']



sub_grade is not included in the model to avoid information leakage

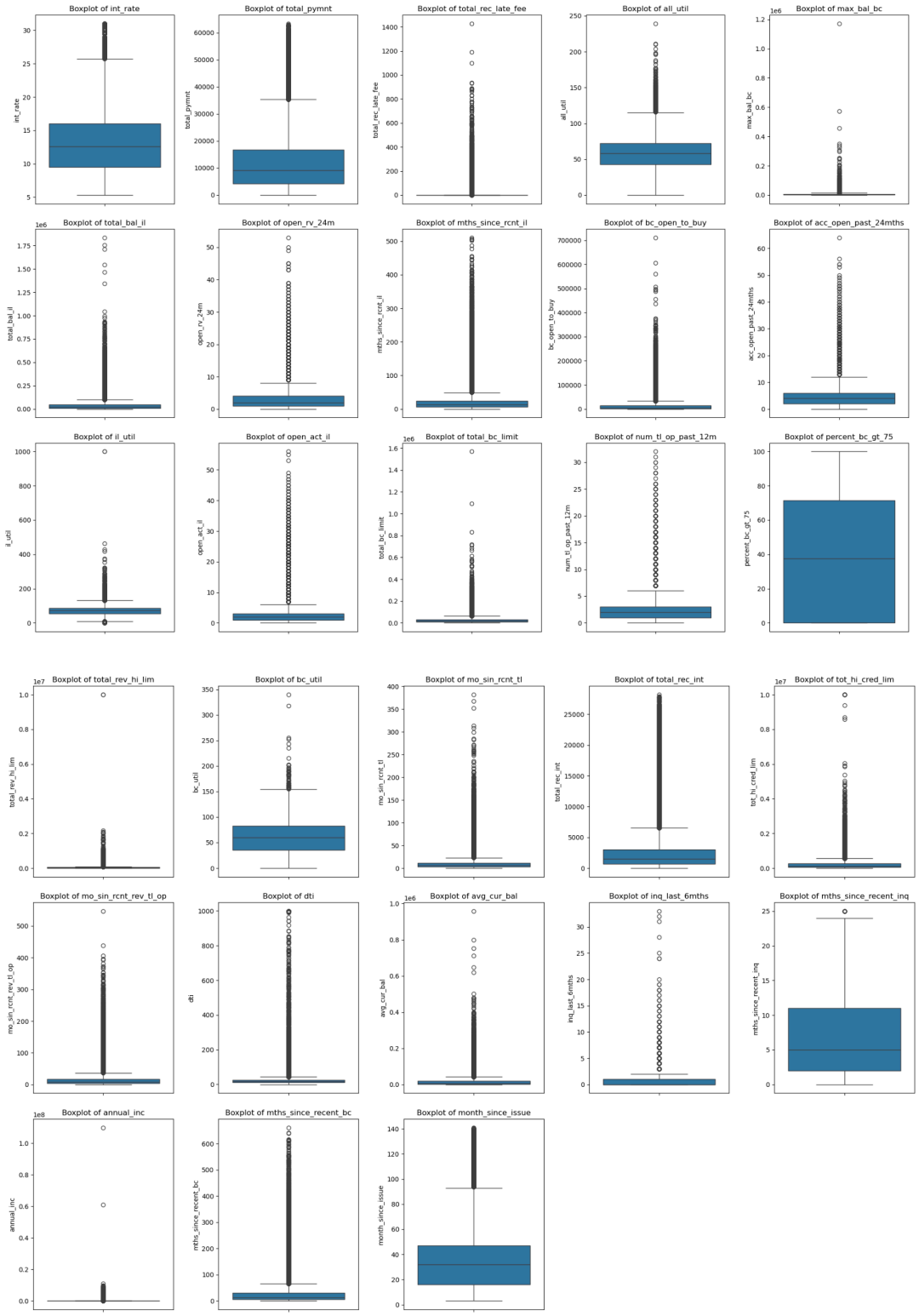
1.2 Preprocess Date Data

Set a reference data and turn them to month_since_ to convert to numeric data type.

1.3 Preprocess Numeric Data

→ ***Multicollinearity Problem:*** Feature pairs (groups) with correlation > 0.85 were identified; the most representative feature from each group (highest sum of pairwise correlations) was retained.

WoE/IV was applied using 10-bin splits (benchmark: 0.03 < IV < 0.5), yielding 28 numeric features. Optimal binning was then determined for each variable using a Decision Tree Classifier to identify split points that maximize separation.



2. Modeling

For target variable, a loan is labelled as default if is has been late for over 90 days or been charged off.

Logistic Regression was trained with L2 regularization (C=10, solver=lbfgs), identified via RandomizedSearchCV. Class imbalance was addressed using `class_weight='balanced'` since resampling techniques (SMOTE, under/oversampling) yielded inferior performance.

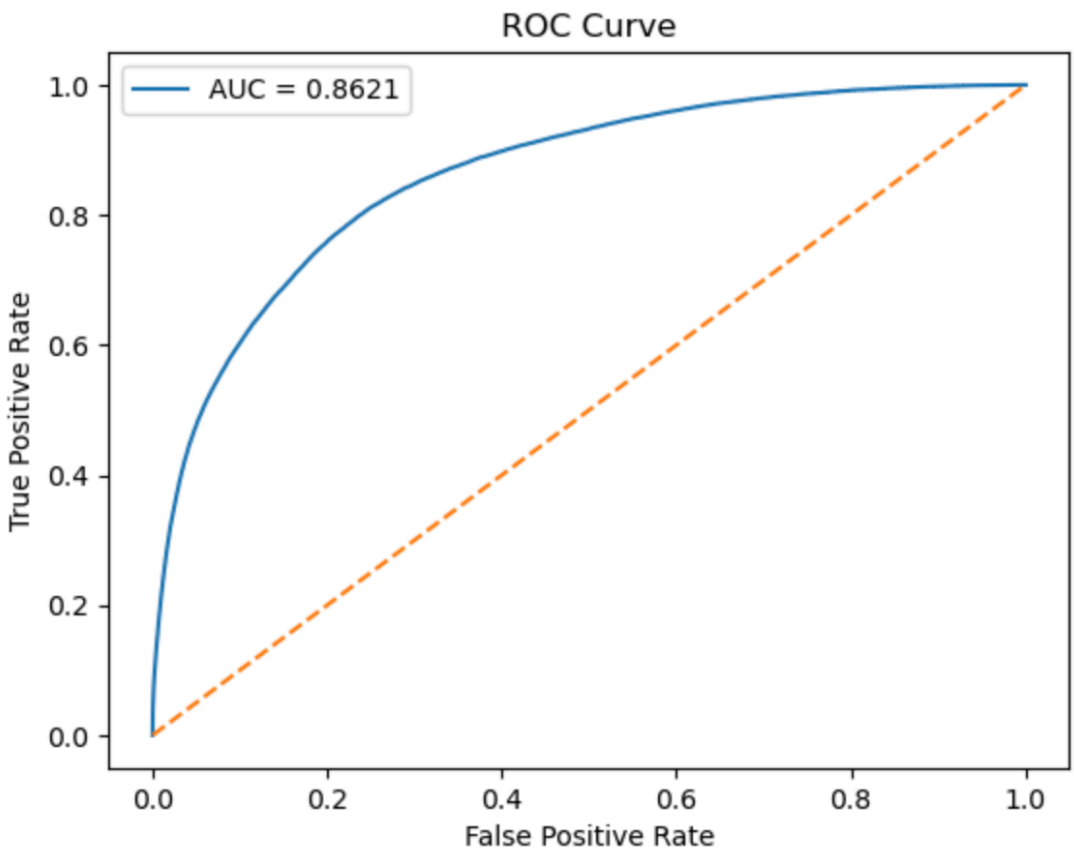
Results: ROC-AUC: 0.862 | KS Statistic: 0.562

At a decision threshold of 0.3: Precision (default class): 0.24 | Recall (default class): 0.90

```
              precision    recall  f1-score   support

           0       0.98      0.59      0.74    394814
           1       0.24      0.90      0.38     56611

    accuracy                           0.63    451425
   macro avg       0.61      0.75      0.56    451425
weighted avg       0.88      0.63      0.69    451425
```



Variable Dictionary

#	Variable Name	Data Type	Description	Category
1	id	string	Unique ID assigned for the specific loan application	Identifiers
2	member_id	string	Unique "Customer Number" (to see if one person has multiple loans)	Identifiers
3	loan_amnt	numeric	The amount the borrower **asked for**	Loan Info
4	funded_amnt	numeric	The amount the bank **approved** and promised to fund	Loan Info
5	funded_amnt_inv	numeric	The portion of the loan actually **funded by investors** (the P2P part)	Loan Info
6	term	categorical	Term of the loan (36 or 60 months)	Loan Info
7	int_rate	numeric	Interest rate on the loan	Loan Info
8	installment	numeric	The **Fixed Monthly Payment** the borrower must pay	Loan Info
9	grade	categorical	Lending Club assigned loan grade (A–G)	Loan Info
10	sub_grade	categorical	Lending Club assigned loan subgrade (A1–G5)	Loan Info
11	emp_title	string	Job title supplied by the borrower	Borrower Profile
12	emp_length	categorical	Employment length in years (0 = <1 year, 10 = 10+ years)	Borrower Profile
13	home_ownership	categorical	Home ownership status: RENT, OWN, MORTGAGE, OTHER	Borrower Profile
14	annual_inc	numeric	Self-reported annual income provided by the borrower	Borrower Profile

#	Variable Name	Data Type	Description	Category
15	verification_status	categorical	Whether income was verified by Lending Club, not verified, or source verified (Lending Club **verified that the source of the income exists**, but they did not necessarily confirm the exact amount of the income)	Borrower Profile
16	issue_d	date	Date the loan was funded	Loan Info
17	loan_status	categorical	Current status of the loan (target variable)	Loan Info
18	pymnt_plan	categorical	Whether a payment plan has been put in place for the loan	Loan Info
19	url	string	URL for the Lending Club page with listing data	Identifiers
20	desc	string	Loan description provided by the borrower (free text)	Borrower Profile
21	purpose	categorical	Borrower-provided category for the loan request	Loan Info
22	title	string	Loan title provided by the borrower (free text)	Loan Info
23	zip_code	string	First 3 digits of the borrower's zip code	Borrower Profile
24	addr_state	categorical	State provided by the borrower in the loan application	Borrower Profile
25	dti	numeric	Debt-to-income ratio (excluding mortgage and Lending Club loan)	Borrower Financials
26	delinq_2yrs	numeric	Number of 30+ day delinquency incidents in the	Credit History

#	Variable Name	Data Type	Description	Category
			borrower's credit file for the past 2 years	
27	earliest_cr_line	date	Date the borrower's earliest reported credit line was opened	Credit History
28	fico_range_low	numeric	Lower boundary of the borrower's **FICO** range at loan origination	Credit History
29	fico_range_high	numeric	Upper boundary of the borrower's **FICO** range at loan origination	Credit History
30	inq_last_6mths	numeric	Number of inquiries in the past 6 months (excluding auto and mortgage)	Credit History
31	mths_since_last_delinq	numeric	Months since the borrower's last delinquency	Credit History
32	mths_since_last_record	numeric	Months since the last public record	Credit History
33	open_acc	numeric	Number of open credit lines in the borrower's credit file (only active account)	Credit History
34	pub_rec	numeric	Number of derogatory public records ***(A public record in a credit context typically refers to legal or government-related filings that suggest financial distress)***	Credit History
35	revol_bal	numeric	Total credit revolving balance	Credit History
36	revol_util	numeric	Revolving line utilization rate (balance / total available revolving credit)	Credit History

#	Variable Name	Data Type	Description	Category
37	total_acc	numeric	Total number of credit lines currently in the borrower's credit file (Every account ever opened)	Credit History
38	initial_list_status	categorical	Initial listing status of the loan (W = whole, F = fractional)	Loan Info
39	out_prncp	numeric	Remaining outstanding principal for total amount funded	Payment History
40	out_prncp_inv	numeric	Remaining outstanding principal for investor-funded portion	Payment History
41	total_pymnt	numeric	Payments received to date for total amount funded	Payment History
42	total_pymnt_inv	numeric	Payments received to date for investor-funded portion	Payment History
43	total_rec_prncp	numeric	Principal received to date	Payment History
44	total_rec_int	numeric	Interest received to date	Payment History
45	total_rec_late_fee	numeric	Late fees received to date	Payment History
46	recoveries	numeric	Post charge-off gross recovery amount	Payment History
47	collection_recovery_fee	numeric	Post charge-off collection fee	Payment History
48	last_pymnt_d	date	Date when last payment was received	Payment History
49	last_pymnt_amnt	numeric	Last total payment amount received	Payment History
50	next_pymnt_d	date	Next scheduled payment date	Payment History
51	last_credit_pull_d	date	The most recent date LendingClub	Credit History

#	Variable Name	Data Type	Description	Category
			pulled the borrower's credit report for this specific loan	
52	last_fico_range_low	numeric	Lower boundary of the borrower's last FICO pulled	Credit History
53	last_fico_range_high	numeric	Upper boundary of the borrower's last FICO pulled ***(* context: Lenders don't just check credit score once. They often pull updates to monitor for signs of distress (like a sudden drop in FICO) while the loan is active)***	Credit History
54	collections_12_mths_ex_med	numeric	Number of collections in 12 months excluding medical collections	Credit History
55	mths_since_last_major_derog	numeric	**Months since the most recent "Major Derogatory" event** (specifically a 90-day or worse rating).	Credit History
56	policy_code	categorical	Publicly available policy_code=1; new products not publicly available policy_code=2	Loan Info
57	application_type	categorical	Whether loan is individual or joint application with co-borrowers	Borrower Profile
58	annual_inc_joint	numeric	Combined self-reported annual income for co-borrowers	Borrower Profile
59	dti_joint	numeric	DTI ratio for co-borrowers (excluding	Borrower Financials

#	Variable Name	Data Type	Description	Category
			mortgages and LC loan)	
60	verification_status_joint	categorical	Whether the co-borrowers' joint income was verified by LC	Borrower Profile
61	acc_now_delinq	numeric	Number of accounts on which the borrower is currently delinquent	Credit History
62	tot_coll_amt	numeric	Total collection amounts ever owed	Credit History
63	tot_cur_bal	numeric	Total current balance of all accounts	Borrower Financials
64	open_acc_6m	numeric	Number of trade lines (total loans) in the last 6 months	Credit History
65	open_act_il	numeric	Number of currently active installment trade lines	Credit History
66	open_il_12m	numeric	Number of installment accounts opened in the past 12 months	Credit History
67	open_il_24m	numeric	Number of installment accounts opened in the past 24 months	Credit History
68	mths_since_rcnt_il	numeric	Months since most recent installment account was opened	Credit History
69	total_bal_il	numeric	Total current balance of all installment accounts	Borrower Financials
70	il_util	numeric	Ratio of total current balance to high credit/limit on all installment accounts	Borrower Financials

#	Variable Name	Data Type	Description	Category
71	open_rv_12m	numeric	Number of revolving trade lines opened in the past 12 months	Credit History
72	open_rv_24m	numeric	Number of revolving trade lines opened in the past 24 months	Credit History
73	max_bal_bc	numeric	highest balance a borrower has ever reached on any of their individual bank-issued credit card	Borrower Financials
74	all_util	numeric	Ratio of balance to credit limit on all trade lines	Borrower Financials
75	total_rev_hi_lim	numeric	Total revolving high credit/credit limit	Borrower Financials
76	inq_fi	numeric	Number of personal finance inquiries	Credit History
77	total_cu_tl	numeric	Number of accounts (trade lines) the borrower has specifically with **Credit Unions**	Credit History
78	inq_last_12m	numeric	Number of credit inquiries in the past 12 months	Credit History
79	acc_open_past_24mths	numeric	Number of trade lines opened in the past 24 months	Credit History
80	avg_cur_bal	numeric	Average current balance of all accounts	Borrower Financials
81	bc_open_to_buy	numeric	Total open-to-buy (available credit) on revolving bankcards	Borrower Financials
82	bc_util	numeric	Ratio of total current balance to credit limit for all bankcard accounts	Borrower Financials
83	chargeoff_within_12_mths	numeric	Number of charge-offs within 12	Credit History

#	Variable Name	Data Type	Description	Category
			months	
84	delinq_amnt	numeric	Past-due amount owed for accounts on which the borrower is currently delinquent	Credit History
85	mo_sin_old_il_acct	numeric	Months since oldest bank installment account was opened	Credit History
86	mo_sin_old_rev_tl_op	numeric	Months since oldest revolving account was opened	Credit History
87	mo_sin_rcnt_rev_tl_op	numeric	Months since most recent revolving account was opened	Credit History
88	mo_sin_rcnt_tl	numeric	Months since most recent account was opened	Credit History
89	mort_acc	numeric	Number of mortgage accounts	Credit History
90	mths_since_recent_bc	numeric	Months since most recent bankcard account was opened	Credit History
91	mths_since_recent_bc_dlq	numeric	Months since most recent bankcard delinquency	Credit History
92	mths_since_recent_inq	numeric	Months since most recent inquiry	Credit History
93	mths_since_recent_revol_delinq	numeric	Months since most recent revolving delinquency	Credit History
94	num_accts_ever_120_pd	numeric	Number of accounts ever 120 or more days past due	Credit History
95	num_actv_bc_tl	numeric	Number of currently active bankcard accounts	Credit History
96	num_actv_rev_tl	numeric	Number of currently active	Credit History

#	Variable Name	Data Type	Description	Category
			revolving trade lines	
97	num_bc_sats	numeric	Number of satisfactory bankcard accounts (***a satisfactory account is one that is not delinquent and is active)***	Credit History
98	num_bc_tl	numeric	Number of bankcard accounts	Credit History
99	num_il_tl	numeric	Number of installment accounts	Credit History
100	num_op_rev_tl	numeric	Number of open revolving accounts	Credit History
101	num_rev_accts	numeric	Number of revolving accounts	Credit History
102	num_rev_tl_bal_gt_0	numeric	Number of revolving trade lines with balance > 0	Credit History
103	num_sats	numeric	Number of satisfactory accounts	Credit History
104	num_tl_120dpd_2m	numeric	Number of accounts currently 120+ days past due (updated in past 2 months)	Credit History
105	num_tl_30dpd	numeric	Number of accounts currently 30 days past due (updated in past 2 months)	Credit History
106	num_tl_90g_dpd_24m	numeric	Number of accounts 90+ days past due in the last 24 months	Credit History
107	num_tl_op_past_12m	numeric	Number of accounts opened in the past 12 months	Credit History
108	pct_tl_nvr_dlq	numeric	Percent of trades never delinquent	Credit History

#	Variable Name	Data Type	Description	Category
109	percent_bc_gt_75	numeric	Percentage of all bankcard accounts > 75% utilization	Borrower Financials
110	pub_rec_bankruptcies	numeric	Number of public record bankruptcies	Credit History
111	tax_liens	numeric	Number of tax liens	Credit History
112	tot_hi_cred_lim	numeric	Total high credit/credit limit across all accounts	Borrower Financials
113	total_bal_ex_mort	numeric	Total credit balance excluding mortgage	Borrower Financials
114	total_bc_limit	numeric	Total bankcard high credit/credit limit	Borrower Financials
115	total_il_high_credit_limit	numeric	Total installment high credit/credit limit	Borrower Financials
116	revol_bal_joint	numeric	Sum of revolving credit balance of co-borrowers, net of duplicate balances	Borrower Financials
117	sec_app_fico_range_low	numeric	FICO range (low) for the secondary applicant	Credit History
118	sec_app_fico_range_high	numeric	FICO range (high) for the secondary applicant	Credit History
119	sec_app_earliest_cr_line	date	Earliest credit line at time of application for the secondary applicant	Credit History
120	sec_app_inq_last_6mths	numeric	Credit inquiries in the last 6 months for the secondary applicant	Credit History
121	sec_app_mort_acc	numeric	Number of mortgage accounts for the secondary applicant	Credit History
122	sec_app_open_acc	numeric	Number of open trades for the	Credit History

#	Variable Name	Data Type	Description	Category
			secondary applicant	
123	sec_app_revol_util	numeric	Revolving utilization ratio for the secondary applicant	Borrower Financials
124	sec_app_open_act_il	numeric	Number of currently active installment trades for the secondary applicant	Credit History
125	sec_app_num_rev_accts	numeric	Number of revolving accounts for the secondary applicant	Credit History
126	sec_app_chargeoff_within_12_mths	numeric	Number of charge-offs within last 12 months for the secondary applicant	Credit History
127	sec_app_collections_12_mths_ex_med	numeric	Number of collections in last 12 months (ex. medical) for the secondary applicant	Credit History
128	sec_app_mths_since_last_major_derog	numeric	Months since most recent 90-day or worse rating for the secondary applicant	Credit History
129	hardship_flag	categorical	Whether the borrower is on a hardship plan	Hardship / Settlement
130	hardship_type	categorical	Type of hardship plan offered	Hardship / Settlement
131	hardship_reason	categorical	Reason the hardship plan was offered	Hardship / Settlement
132	hardship_status	categorical	Status of the hardship plan (active, pending, canceled, completed, broken)	Hardship / Settlement
133	deferral_term	numeric	Number of months the borrower is expected to pay	Hardship / Settlement

#	Variable Name	Data Type	Description	Category
			less due to hardship plan	
134	hardship_amount	numeric	Interest payment the borrower committed to making monthly while on hardship plan	Hardship / Settlement
135	hardship_start_date	date	Start date of the hardship plan period	Hardship / Settlement
136	hardship_end_date	date	End date of the hardship plan period	Hardship / Settlement
137	payment_plan_start_date	date	Date the first hardship plan payment is due	Hardship / Settlement
138	hardship_length	numeric	Number of months the borrower will make smaller payments due to hardship plan	Hardship / Settlement
139	hardship_dpd	numeric	Account days past due as of the hardship plan start date	Hardship / Settlement
140	hardship_loan_status	categorical	Loan status as of the hardship plan start date	Hardship / Settlement
141	orig_projected_additional_accrued_interest	numeric	Original projected additional interest to accrue for the hardship payment plan	Hardship / Settlement
142	hardship_payoff_balance_amount	numeric	Payoff balance amount as of the hardship plan start date	Hardship / Settlement
143	hardship_last_payment_amount	numeric	Last payment amount as of the hardship plan start date	Hardship / Settlement
144	disbursement_method	categorical	Method by which the borrower receives their loan (CASH or DIRECT_PAY)	Loan Info

#	Variable Name	Data Type	Description	Category
145	debt_settlement_flag	categorical	Whether the charged-off borrower is working with a debt settlement company	Hardship / Settlement